



SWINBURNE
UNIVERSITY OF
TECHNOLOGY

COS10004 Computer Systems

Lecture 12.1 Final Assessment

CRICOS provider 00111D

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COS10004 assessment

- Labs: 10%
- Assignments (15%, 15%)
- Final Assessment 60%

Final Assessment

- An online Quiz on Canvas.
- An announcement will provide the link on the day

Day and Date	Thursday 12th November 2020
Time of Assessment	9am AEST (Melbourne)
Duration of assessment window	24 hours
Other instructions	This assessment will be available in Canvas only for the duration window noted above. Students must submit their work <u>8:59am Friday 13th November</u>

Final Assessment

- Total of 36 questions:
 - MCQs (~10%)
 - short answer questions (~20%)
 - medium answer questions (~20%)
 - large questions (~10%)
- Only a single attempt allowed
- Should take no more than 3 hours to complete

Final Assessment – Multiple Choice

- 10 MCQs
 - Select the 1 best answer
 - 1 mark each
 - Similar style to the sample quiz questions
 - based on Weeks 1-11 (minus lectures 10.3 and 10.4 in Week 11)
- Logic gates, circuits, memory, RPi, ASM

topics

- Shift register - what it does to a binary number
- AND gates, OR gates in circuits
- Adders, half adders
- Number conversions
- ASM indirect addressing
- mov and ldr
- RPI hardware
- counters
- Memory types
- Data communication

Short answer questions

- 16 short questions – 1-4 marks each,
- 1-line answers (text) up to 1/2 page answers

topics

- Number (radix) conversions
- Data communication (serial versus parallel, RS-232)
- Computing architectures (Von Neumann versus Harvard)
- ARM assembly instructions
- Stacks (hardware and software)
- Memory types
- Register vs. shift register
- Asynchronous vs. synchronous counters (incl. MOD)
- Interrupts vs. polling
- Encoders and decoders
- Stacks
- Adders

Medium questions

- 6 questions
- 3-4 marks each
- Topics:
 - Number representations (BCD, IEEE745, Fixed point representation, 2s compliment)
 - RS232 serial data communications
 - Reading and interpreting assembly code
 - RISC vs. CISC

Extended questions

- 4 questions
- Up to 5 marks
- Topics:
 - Reading assembly code
 - what each line does
 - what all of it does
 - Writing ASM
 - functions
 - Microprocessors versus microcontrollers

Recurring themes

- Board selection
- Binary arithmetic
- Algorithms for arithmetic
- Number representation
- Block-level circuits (registers, counters...)
- Assembly
- How computers work

Final Assessment Conditions

- Final assessment is not intended to be open book, however this is clearly not enforceable
- Expectations therefore are:
 - you can refer to lecture notes, however this will be of only limited value:
 - Much will rely on your overall familiarity and ability to apply the knowledge from lectures and labs
 - You will work alone – no communication should occur
 - You will not copy online sources.
 - We are very aware of what resources exist online and will be looking closely at answers.
 - A post assessment interview may be requested if doubt exists

Things to do:

- Assignment 2
 - really good revision
 - do the same with the logic circuits.
- Do the:
 - sample exam (soon to be released)
 - online quizzes (more to come soon)
 - the labs. Properly this time.
- Go through the lectures - Summarise as you go.
- Draw diagrams
 - circuits
 - data types
 - how computers work
- Pretend to compile a cheat sheet.
 - Helps you review, revise/organise your knowledge.
 - Test it (on the practice exams)

Summary

- Final Assessment will cover all topics except screen graphics and ASM in an OS (Lectures 10.3 and 10.4)
- Worth 60%
- Thursday November 12, from 9am
- Online quiz available for 24 hours